

Section 4: Physical chemistry

- a) Acids, alkalis and salts
- b) Energetics
- c) Rates of reaction
- d) Equilibria

- a) Acids, alkalis and salts

Students will be assessed on their ability to:

- 4.1 describe the use of the indicators litmus, phenolphthalein and methyl orange to distinguish between acidic and alkaline solutions
- 4.2 understand how the pH scale, from 0–14, can be used to classify solutions as strongly acidic, weakly acidic, neutral, weakly alkaline or strongly alkaline
- 4.3 describe the use of universal indicator to measure the approximate pH value of a solution
- 4.4 define acids as sources of hydrogen ions, H^+ , and alkalis as sources of hydroxide ions, OH^-
- 4.5 predict the products of reactions between dilute hydrochloric, nitric and sulfuric acids; and metals, metal oxides and metal carbonates (excluding the reactions between nitric acid and metals)
- 4.6 recall the general rules for predicting the solubility of salts in water:
 - i all common sodium, potassium and ammonium salts are soluble
 - ii all nitrates are soluble
 - iii common chlorides are soluble, except silver chloride
 - iv common sulfates are soluble, except those of barium and calcium
 - v common carbonates are insoluble, except those of sodium, potassium and ammonium
- 4.7 describe how to prepare soluble salts from acids
- 4.8 describe how to prepare insoluble salts using precipitation reactions
- 4.9 describe how to carry out acid-alkali titrations.

b) Energetics

Students will be assessed on their ability to:

- 4.10 recall that chemical reactions in which heat energy is given out are described as exothermic and those in which heat energy is taken in are endothermic
- 4.11 describe simple calorimetry experiments for reactions such as combustion, displacement, dissolving and neutralisation in which heat energy changes can be calculated from measured temperature changes
- 4.12 calculate molar enthalpy change from heat energy change**
- 4.13 understand the use of ΔH to represent molar enthalpy change for exothermic and endothermic reactions
- 4.14 represent exothermic and endothermic reactions on a simple energy level diagram
- 4.15 recall that the breaking of bonds is endothermic and that the making of bonds is exothermic